

Factorizations for Data Analysis: First Examples

Module 1 of 4

Applied Math 210: Algebraic Fundamentals of Representing Data

Anna Seigal

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Overview

Across scientific domains, researchers seek to describe and understand complex, high-dimensional systems, such as a genome, a brain, a corpus of texts, or an AI model. A central guiding principle is that of compositionality: complex systems can be described as a combination of smaller, simpler parts. Algebraic Fundamentals of Representing Data (AM210) is a course that studies what these parts should be, and how to combine them, starting from the first classical paradigms for data analysis. We formalize the idea of expressing a complex system in terms of simpler parts as a *factorization*.

The course is in four parts. We begin, in part 1, with factorizations of matrices. Part 2 studies tensors, the multilinear extension of a matrix that arises when searching for higher-order structure in data and for data that is organized by more than two indices, such as variables observed across multiple contexts. Part 3 studies graphical models and causal inference, encoding nodes as variables and the edges between them as their dependencies. Part 4 considers deep learning models, where complex functions are broken into simpler parts via composition.

Three questions guide the course:

- **Expressivity.** What can a model represent? When does a factorization exist?
- **Identifiability.** Is a factorization unique?
- **Optimization landscapes.** How do you compute a factorization? What do the optimization problem, and its solution, look like?

This document contains the lecture notes for the first part of the course, from Fall 2025.

Introduction

Compositionality of complex systems is the hypothesis that observed variables can be described as combinations of (usually unobserved) simpler parts. For example, gene expression patterns can be encoded via gene modules, brain data via neural sources, and texts via their concepts and topics.

Formalized via random vectors, this hypothesis writes an observed random vector $\mathbf{x} \in \mathbb{R}^p$ as $\mathbf{x} = f(\mathbf{z})$ for a latent random vector $\mathbf{z} \in \mathbb{R}^q$, and a fixed unknown function $f : \mathbb{R}^q \rightarrow \mathbb{R}^p$. The special case where the combinations are linear replaces the map f by a fixed, unknown matrix $A \in \mathbb{R}^{p \times q}$. Recovering f or A and properties of $\mathbf{z}$ from samples of $\mathbf{x}$ is a factorization problem.

The first part of the course studies the first, and most classical, example: uncorrelated latent variables $\mathbf{z}$ transformed via an orthogonal transformation. This is principal component analysis

(PCA). We use the spectral theorem and Eckart–Young theorem to derive the classical result that the principal components are the eigenvectors of the sample covariance matrix.

PCA is the starting point for many other factorizations, including sparse coding and dictionary learning (linear in the combination rule with a sparsity constraint on $\mathbf{z}$), autoencoders (a linear autoencoder recovers PCA), and learned representations of deep networks (a direction in a network’s activation space generalizes a principal component).

Useful texts

This module of the course studies the linear algebra behind factorizations for data analysis. Each linear algebra result we study is motivated by a question about representing data. Linear algebra texts (Strang, *Linear Algebra and Learning from Data*; Horn & Johnson, *Matrix Analysis*) derive the spectral theorem and the SVD, without as much focus on covariance matrices or PCA; applied statistics and machine learning texts (e.g. Bishop, *Pattern Recognition and Machine Learning*) focus on PCA as a tool, without as much emphasis on proofs. The course sits in between these two styles. A related reference is my new article “Factorizations for Data Analysis,” *SIAM News*, Vol. 58, No. 7, 2025, which covers a few key factorizations in brief.

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1 Notation

We use bold lowercase for vectors, both fixed and random, plain lowercase for scalars, plain uppercase for matrices. A subscript on a bold symbol labels a vector (e.g. $\mathbf{v}_i$, the i -th eigenvector), while the i th entry of a vector $\mathbf{v}$ is denoted v_i (without bold).

Symbol	Meaning
$\mathbf{x} = (x_1, \dots, x_p)^\top$	random vector of p observed variables
$\mathbf{z} = (z_1, \dots, z_q)^\top$	random vector of q latent variables
$\mathbf{x}^{(i)} \in \mathbb{R}^p$	the i -th sample: row i of X , a realization of $\mathbf{x}$
$X \in \mathbb{R}^{n \times p}$	data matrix: n samples (rows), p variables (columns); row i is $(\mathbf{x}^{(i)})^\top$
$Z \in \mathbb{R}^{n \times q}$	scores matrix; column k is module k 's per-sample expression
$A \in \mathbb{R}^{p \times q}$	mixing matrix, $\mathbf{x} = A\mathbf{z}$; column k is module k 's per-variable loading
$\mathbf{u}, \mathbf{v}_i \in \mathbb{R}^p$	vectors
$\Sigma_{\mathbf{x}}, \Sigma_{\mathbf{z}}$	population covariance of $\mathbf{x}$ and $\mathbf{z}$, respectively
$S \in \mathbb{R}^{p \times p}$	sample covariance matrix
$S^2(\mathbb{R}^p)$	the space of $p \times p$ symmetric matrices
$O(p)$	the space of $p \times p$ orthogonal matrices
Diag_p	the space of $p \times p$ diagonal matrices
$S = V\Lambda V^\top$	eigendecomposition: V orthogonal, $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_p)$, $\lambda_1 \geq \dots \geq \lambda_p \geq 0$
S_r	truncation of S to the top r eigenvectors and eigenvalues
$M = UDV^\top$	SVD of a matrix $M \in \mathbb{R}^{n \times p}$; $D = \text{diag}(\sigma_1, \dots, \sigma_{\min(n,p)})$, its singular values
PC_i	i -th principal component direction

2 Data as a Matrix

Multivariate data classically consists of a collection of vectors $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)} \in \mathbb{R}^p$. Each sample vector $\mathbf{x}^{(i)} \in \mathbb{R}^p$ gives the values of p variables. Such data can be organized as a matrix $X \in \mathbb{R}^{n \times p}$: n samples, p variables, entry X_{ij} is the value of variable j in sample i . For example:

Variable j	Sample i	X_{ij}
gene	cell	expression level of gene j in cell i
electrode	timepoint	voltage of electrode j at time i
pixel	image	intensity of pixel j in image i
word	document	count of word j in document i

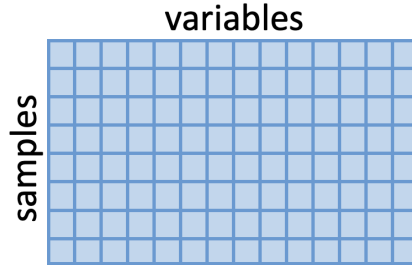


Figure 1: A data matrix, samples as rows, variables as columns

Data analysis seeks to uncover the building blocks that describe and explain such a matrix. Doing so amounts to fitting a model to the data. We view the vectors $\mathbf{x}^{(i)}, i = 1, \dots, n$ as samples of a vector of observed variables $\mathbf{x} = (x_1, \dots, x_p)^\top$. Writing observed variables as a linear combination of parts writes $\mathbf{x} = A\mathbf{z}$. We call A the mixing matrix: it encodes how the latent variables mix

together to form the observed variables. At the level of samples, this says $\mathbf{x}^{(i)} = A\mathbf{z}^{(i)}$. Collecting together across all samples, this yields the matrix equation

$$X = ZA^\top,$$

where $X \in \mathbb{R}^{n \times p}$ is the data matrix and $Z \in \mathbb{R}^{n \times q}$ is the scores matrix. This product can also be written as a sum

$$X = \sum_{k=1}^q \mathbf{z}_k \mathbf{a}_k^\top,$$

where $\mathbf{z}_k$ and $\mathbf{a}_k$ are the k th columns of Z and A , respectively. The p observed variables have been expressed via q modules.

Definition 2.1 (Module). A *module* is a rank-one piece of the data matrix, $\mathbf{z}\mathbf{a}^\top$. It is built from a *score* $\mathbf{z} \in \mathbb{R}^n$ (how much the pattern is expressed in each sample) and a *loading* $\mathbf{a} \in \mathbb{R}^p$ (how much each variable participates in it).

Various choices can be made for the structure to impose on the collection $\{(\mathbf{z}_k, \mathbf{a}_k) : k \in [q]\}$, such as orthogonality, sparsity, etc. For $q < p$, the decomposition is a low-rank approximation of X .

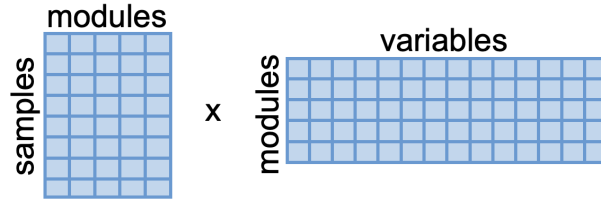


Figure 2: A low-rank decomposition $X = ZA^\top$ of a data matrix writes it in terms of modules: the columns of Z record how much the module is expressed in each sample, while the rows of $A^\top$ record how much each observed variable participates in it. In this example, there are $n = 8$ samples, $p = 14$ variables, and $q = 5$ modules.

3 Linear Latent-Variable Models

Definition 3.1 (Linear latent-variable model). Let $\mathbf{z} \in \mathbb{R}^q$ be a random vector of latent (unobserved) variables and $A \in \mathbb{R}^{p \times q}$ a fixed, unknown matrix. The observed random vector is

$$\mathbf{x} = A\mathbf{z}.$$

Proposition 3.2 (Mean and covariance under a linear map). *Let $\mathbf{z}$ have mean $\mu \in \mathbb{R}^q$ and covariance $\Sigma_{\mathbf{z}} = \mathbb{E}[(\mathbf{z} - \mu)(\mathbf{z} - \mu)^\top] \in \mathbb{R}^{q \times q}$. Then $\mathbf{x} = A\mathbf{z}$ has mean $A\mu$ and covariance $A\Sigma_{\mathbf{z}}A^\top$.*

Proof. By linearity of expectation, $\mathbb{E}[A\mathbf{z}]_i = \mathbb{E}[\sum_j A_{ij}z_j] = \sum_j A_{ij}\mathbb{E}[z_j] = (A\mu)_i$. For the covariance,

$$\text{Cov}(A\mathbf{z}) = \mathbb{E}[(A\mathbf{z} - A\mu)(A\mathbf{z} - A\mu)^\top] = \mathbb{E}[A(\mathbf{z} - \mu)(\mathbf{z} - \mu)^\top A^\top] = A \mathbb{E}[(\mathbf{z} - \mu)(\mathbf{z} - \mu)^\top] A^\top = A\Sigma_{\mathbf{z}}A^\top,$$

again by linearity of expectation, which extracts the constant matrices A , $A^\top$. $\square$

Remark 3.3 (The factorization problem). Proposition 3.2 gives $\Sigma_{\mathbf{x}} = A\Sigma_{\mathbf{z}}A^\top$. In practice $\Sigma_{\mathbf{x}}$ is known (or estimated), and $A, \Sigma_{\mathbf{z}}$ are unknown. Recovering A and $\Sigma_{\mathbf{z}}$ from (an estimate of) $\Sigma_{\mathbf{x}}$ is a matrix factorization problem. For invertible $A \in \mathbb{R}^{q \times q}$, $\Sigma \mapsto A\Sigma A^\top$ is the action of $GL(q)$ on symmetric matrices by congruence. The output of a factorization problem is an estimate for the factors: here, this is the covariance matrix $\Sigma_{\mathbf{z}}$ and the mixing matrix A . Provided A has a left-inverse, one can use the relationship $AZ^\top = X^\top$ to obtain the scores matrix Z .

4 Symmetric and Positive Semi-Definite Matrices

Definition 4.1. $S \in \mathbb{R}^{p \times p}$ is *symmetric* if $S = S^\top$. It is *positive semi-definite* (PSD, denoted $S \succeq 0$) if it is symmetric and $\mathbf{v}^\top S \mathbf{v} \geq 0$ for all $\mathbf{v} \in \mathbb{R}^p$. It is *positive definite* (PD, denoted $S \succ 0$) if $\mathbf{v}^\top S \mathbf{v} > 0$ for all $\mathbf{v} \neq 0$.

Proposition 4.2. $\Sigma_{\mathbf{x}} = \text{Cov}(\mathbf{x})$ is PSD.

Proof. Assume without loss of generality that $\mathbf{x}$ is mean-centered ($\mu = 0$), so $(\Sigma_{\mathbf{x}})_{ij} = \mathbb{E}[x_i x_j]$. Symmetry follows from the fact that $\mathbb{E}[x_i x_j] = \mathbb{E}[x_j x_i]$. To show that $\Sigma_{\mathbf{x}}$ is PSD: let $\mathbf{v} \in \mathbb{R}^p$. The linear combination of variables $\mathbf{v}^\top \mathbf{x} = \sum_i v_i x_i$ is a scalar random variable. Then

$$\mathbf{v}^\top \Sigma_{\mathbf{x}} \mathbf{v} = \mathbf{v}^\top \mathbb{E}[\mathbf{x} \mathbf{x}^\top] \mathbf{v} = \mathbb{E}[(\mathbf{v}^\top \mathbf{x})^2] = \text{Var}(\mathbf{v}^\top \mathbf{x}) \geq 0,$$

using $\mathbb{E}[x_i^2] = \text{Var}(x_i)$ for the mean-centered $\mathbf{v}^\top \mathbf{x}$, and the fact that variance is non-negative. $\square$

Definition 4.3 (Sample covariance). Given data $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)} \in \mathbb{R}^p$ with sample mean $\bar{\mathbf{x}} = \frac{1}{n} \sum_i \mathbf{x}^{(i)}$, the *sample covariance matrix* is

$$S = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}^{(i)} - \bar{\mathbf{x}})(\mathbf{x}^{(i)} - \bar{\mathbf{x}})^\top.$$

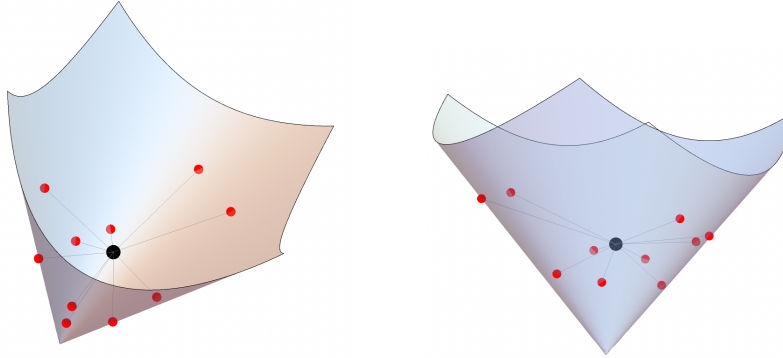


Figure 3: The 2×2 PSD cone (two views). Its boundary consists of matrices $\mathbf{v}\mathbf{v}^\top$ for $\mathbf{v} \in \mathbb{R}^2$. Here 10 mean-centered points (say $\mathbf{v}\mathbf{v}^\top$ for $\mathbf{v} \in \{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)}\}$) on the boundary are shown in red. Their average (in black) is the sample covariance matrix of the data $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)}$.

In practice the true (population) $\Sigma_{\mathbf{x}}$ is unknown, and S estimates it.

Exercise 4.4. The sample covariance matrix S is PSD. When is it PD?

5 The Spectral Theorem

Definition 5.1. $V \in \mathbb{R}^{p \times p}$ is *orthogonal* if $VV^\top = I$. $D \in \mathbb{R}^{p \times p}$ is *diagonal* if $D_{ij} = 0$ for $i \neq j$.

Theorem 5.2 (Spectral theorem, or eigendecomposition). *A real symmetric matrix $S \in \mathbb{R}^{p \times p}$ can be written $S = V\Lambda V^\top$ with V orthogonal and Λ diagonal. If the diagonal entries of Λ (the eigenvalues of S) are distinct and in descending order, the factorization is unique up to the signs of the columns of V .*

The proof has two parts: showing that if such a decomposition holds, its pieces are eigenpairs of S ; and showing such a decomposition always exists. We begin with the first part.

Proposition 5.3. *If $S = V\Lambda V^\top$ with $V = [\mathbf{v}_1 \cdots \mathbf{v}_p]$ orthogonal and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_p)$, then $S\mathbf{v}_i = \lambda_i \mathbf{v}_i$ for each i .*

Proof. $S\mathbf{v}_i = V\Lambda V^\top \mathbf{v}_i = V\Lambda \mathbf{e}_i$, since $V^\top \mathbf{v}_i = \mathbf{e}_i$ (V orthogonal, so its columns are orthonormal). Then $\Lambda \mathbf{e}_i = \lambda_i \mathbf{e}_i$, so $S\mathbf{v}_i = V(\lambda_i \mathbf{e}_i) = \lambda_i V \mathbf{e}_i = \lambda_i \mathbf{v}_i$. $\square$

Exercise 5.4. If $\mathbf{v}_i, \mathbf{v}_j$ are eigenvectors of S with distinct eigenvalues $\lambda_i \neq \lambda_j$, then $\mathbf{v}_i \perp \mathbf{v}_j$.

A generic S has p distinct eigenvalues, since these are the roots of $\det(S - \lambda I) = 0$, a degree- p polynomial. These give a full orthogonal basis of eigenvectors. Existence of a full *orthogonal* eigenbasis when eigenvalues repeat is more subtle (extending an orthogonal set of eigenvectors to a full orthogonal basis when an eigenspace has dimension > 1); we do not prove it here — see Horn & Johnson, *Matrix Analysis*.

For uniqueness, we observe that in the case of distinct eigenvalues each eigenspace is one-dimensional, so there is only one representative vector $\mathbf{v}_i$ up to sign. Once the eigenvalues are listed in descending order, this fixes the matrices V and Λ , up to sign flips on the columns of V .

What Parameter Counting Does and Does Not Show

$S = V\Lambda V^\top$ is a factorization of a symmetric matrix. To help gain insight into its existence and uniqueness, we can check that both sides have the same number of free parameters:

$$\dim(S^2(\mathbb{R}^p)) = \binom{p+1}{2}, \quad \dim(O(p)) = \binom{p}{2}, \quad \dim(\text{Diag}_p) = p,$$

where $\dim(O(p))$ counts the dimension of the orthogonal group as a manifold. Since $\binom{p}{2} + p = \binom{p+1}{2}$, the counts match.

Remark 5.5. A matching parameter count proves neither existence nor uniqueness. But a mismatch is informative: too few free parameters on the right rules out existence in general (there is no room to represent every symmetric matrix this way), while too many contradicts uniqueness (there is more than one way to write the same matrix, generically). Matching counts is necessary for existence and uniqueness to be plausible.

Corollary 5.6. *The spectral theorem writes the observed variables $\mathbf{x}$ as an orthogonal transformation ($A = V$) of p uncorrelated latent variables $\mathbf{z}$ ($\Sigma_{\mathbf{z}} = \Lambda$ diagonal).*

6 Rank and Eckart–Young

Definition 6.1 (Rank). A matrix $M \in \mathbb{R}^{n \times p}$ has *rank 1* if $M = \lambda \mathbf{u} \mathbf{v}^\top$ for some $\lambda \in \mathbb{R}$, $\mathbf{u} \in \mathbb{R}^n$, $\mathbf{v} \in \mathbb{R}^p$ (both non-zero). It has *rank r* if it can be written $M = \sum_{i=1}^r \lambda_i \mathbf{u}_i \mathbf{v}_i^\top$ for some $\lambda_i \in \mathbb{R}$, $\mathbf{u}_i \in \mathbb{R}^n$, $\mathbf{v}_i \in \mathbb{R}^p$, and r is the smallest number of terms for which such a sum exists. A symmetric $M \in S^2(\mathbb{R}^p)$ has rank r if $M = \sum_{i=1}^r \lambda_i \mathbf{v}_i \mathbf{v}_i^\top$ (same $\mathbf{v}_i$ on both sides), and not fewer.

Lemma 6.2 (Product form $\Leftrightarrow$ sum-of-rank-one form). Let $A \in \mathbb{R}^{p \times r}$ have columns $\mathbf{a}_1, \dots, \mathbf{a}_r$ and $B \in \mathbb{R}^{n \times r}$ have columns $\mathbf{b}_1, \dots, \mathbf{b}_r$. Then

$$AB^\top = \sum_{i=1}^r \mathbf{a}_i \mathbf{b}_i^\top.$$

Proof. Entrywise, $(AB^\top)_{jk} = \sum_{i=1}^r A_{ji} B_{ki} = \sum_{i=1}^r (\mathbf{a}_i)_j (\mathbf{b}_i)_k = \sum_{i=1}^r (\mathbf{a}_i \mathbf{b}_i^\top)_{jk}$. $\square$

Lemma 6.2 gives the equivalent forms $M = U \Lambda V^\top$ (product) and $M = \sum_i \lambda_i \mathbf{u}_i \mathbf{v}_i^\top$ (sum). They are used interchangeably from here on. Both are useful perspectives for data analysis and generalizations beyond PCA.

Definition 6.3 (Frobenius inner product). For $A, B \in \mathbb{R}^{n \times p}$, $\langle A, B \rangle = \sum_{ij} A_{ij} B_{ij} = \text{tr}(AB^\top)$, and $\|A\|^2 = \langle A, A \rangle$.

Two facts used below: $\text{tr}(AB) = \text{tr}(BA)$ whenever both products are defined, and, for a vector $\mathbf{u} \in \mathbb{R}^p$, $\langle S, \mathbf{u} \mathbf{u}^\top \rangle = \mathbf{u}^\top S \mathbf{u}$ and $\langle \mathbf{u} \mathbf{u}^\top, \mathbf{u} \mathbf{u}^\top \rangle = \|\mathbf{u}\|^4$.

Theorem 6.4 (Eckart–Young, symmetric case). Let $S = V \Lambda V^\top \in S^2(\mathbb{R}^p)$, $V = [\mathbf{v}_1 \cdots \mathbf{v}_p]$, $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_p)$, $\lambda_1 \geq \cdots \geq \lambda_p \geq 0$. Let $V_r = [\mathbf{v}_1 \cdots \mathbf{v}_r]$, $\Lambda_r = \text{diag}(\lambda_1, \dots, \lambda_r)$, and $S_r = V_r \Lambda_r V_r^\top$. Then for any M with $\text{rank}(M) \leq r$,

$$\|S - M\| \geq \|S - S_r\|.$$

Proof. We prove the result in three steps.

Step 1: S and M both diagonal. Write $S = D = \text{diag}(d_1, \dots, d_p)$, $M = \Lambda' = \text{diag}(\lambda'_1, \dots, \lambda'_p)$ with at most r non-zero entries, at indices $I \subseteq \{1, \dots, p\}$, $|I| = r$. Then

$$\|D - \Lambda'\|^2 = \sum_{i \in I} (d_i - \lambda'_i)^2 + \sum_{i \notin I} d_i^2.$$

The first sum is minimized (at 0) by $\lambda'_i = d_i$; the second is smallest when I picks out the r largest $|d_i|$, i.e. $I = \{1, \dots, r\}$ given the ordering $d_1 \geq \cdots \geq d_p \geq 0$. So the best diagonal rank- r approximation of D is $D_r = \text{diag}(d_1, \dots, d_r, 0, \dots, 0)$.

Step 2: $S = D$ diagonal, M general. Write $\|D - M\|^2 = \|D\|^2 + \|M\|^2 - 2 \text{tr}(DM)$. Let $\mu_1 \geq \cdots \geq \mu_p \geq 0$ be the eigenvalues of M (rank $\leq r$, so at most r are non-zero). Von Neumann's trace inequality states

$$\text{tr}(DM) \leq \sum_{i=1}^p d_i \mu_i$$

(we use, but do not reprove, this fact; see Horn & Johnson, *Matrix Analysis*). Hence

$$\|D - M\|^2 \geq \sum_i d_i^2 + \sum_i \mu_i^2 - 2 \sum_i d_i \mu_i = \sum_{i=1}^r (d_i - \mu_i)^2 + \sum_{i>r} d_i^2 \geq \sum_{i>r} d_i^2 = \|D - D_r\|^2,$$

using $\mu_i = 0$ for $i > r$, and Step 1's conclusion that the middle sum is minimized at $\mu_i = d_i$ for $i \leq r$. So D_r beats every rank- $\leq r$ M , not just diagonal ones.

Step 3: general S . Write $S = VDV^\top$. For any rank- r M , $V^\top MV$ also has rank r (V invertible), and an orthogonal change of basis preserves the Frobenius norm ($\|VNV^\top\|^2 = \text{tr}(VNV^\top VNV^\top) = \text{tr}(NN^\top) = \|N\|^2$). So

$$\|S - M\| = \|V(D - V^\top MV)V^\top\| = \|D - V^\top MV\| \geq \|D - D_r\| = \|S - S_r\|,$$

the middle inequality by Step 2 applied to the rank- r matrix $V^\top MV$. $\square$

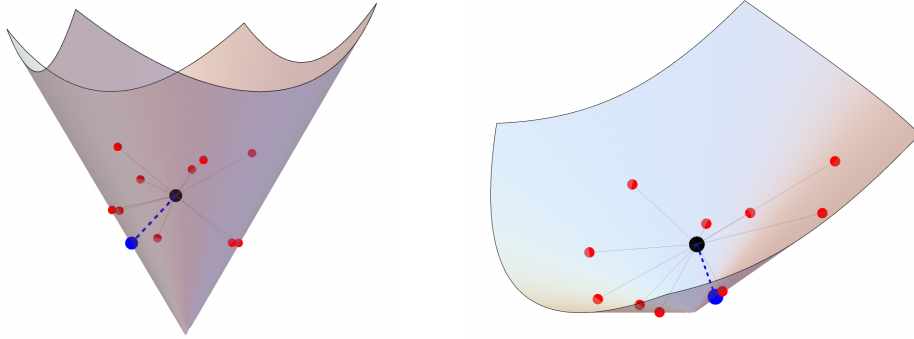


Figure 4: The same 2×2 PSD cone as above in Figure 3 (two views), with the same red points and their average in black. The projection of S to the boundary is in blue. This is the best rank one approximation, which is the top principal component of the red data.

Remark 6.5 (Beyond the Frobenius norm). Theorem 6.4 holds for any unitarily invariant norm, such as the spectral norm, the nuclear norm, or a Schatten- p norm.

Remark 6.6 (Beyond PSD matrices). Positive semi-definiteness of S is not required here: the spectral theorem holds for any real symmetric matrix, and so does Theorem 6.4, provided the ordering and truncation are by $|\lambda_i|$ rather than λ_i (the r eigenpairs kept are the ones of largest magnitude). The PSD property (all $\lambda_i \geq 0$) is a property of covariance matrices, not a requirement for diagonalization.

7 Principal Component Analysis (PCA)

Consider data $X \in \mathbb{R}^{n \times p}$, mean-centered, with sample covariance $S = \frac{1}{n}X^\top X$.

Definition 7.1 (Principal components). The i -th *principal component* PC_i is a unit direction $\mathbf{u} \in \mathbb{R}^p$ maximizing the variance $\mathbf{u}^\top S \mathbf{u}$ of the data projected onto $\mathbf{u}$, subject to $\mathbf{u} \perp \text{PC}_1, \dots, \text{PC}_{i-1}$.

Definition 7.2 (Total variance in a subspace). For a subspace $U \subseteq \mathbb{R}^p$ with orthonormal basis $\mathbf{u}_1, \dots, \mathbf{u}_r$, the total variance in U is $\sum_{i=1}^r \mathbf{u}_i^\top S \mathbf{u}_i = \text{tr}(U^\top S U)$, $U = [\mathbf{u}_1 \cdots \mathbf{u}_r]$.

PCA finds the r -dimensional subspace maximizing total variance: U maximizing $\text{tr}(U^\top S U)$. Any orthonormal basis of the maximizing subspace achieves the maximum, so a maximizer is unique up to choice of basis of $\text{span}\{\text{PC}_1, \dots, \text{PC}_r\}$. Choosing a basis by extending $\text{span}\{\text{PC}_1, \dots, \text{PC}_{i-1}\}$, as i varies, makes principal components generically unique.

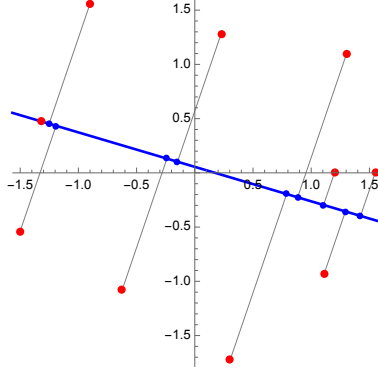


Figure 5: A 2D synthetic data cloud with PC_1 marked in blue. This is the dataset visualized as points on the boundary of the PSD cone in Figure 4. The blue line is $\text{Span}(\mathbf{v})$. The blue dot in Figure 4 is the point $\mathbf{v}\mathbf{v}^\top$.

Proposition 7.3. $PC_1 = \mathbf{v}_1$, the top eigenvector of S .

Proof. Maximize $\mathbf{u}^\top S \mathbf{u}$ subject to $\|\mathbf{u}\|^2 = 1$. The Lagrangian is $L(\mathbf{u}, \lambda) = \mathbf{u}^\top S \mathbf{u} - \lambda(\|\mathbf{u}\|^2 - 1)$, and

$$\frac{\partial L}{\partial u_i} = 2(S\mathbf{u})_i - 2\lambda u_i = 0 \quad \forall i \implies S\mathbf{u} = \lambda \mathbf{u} :$$

$\mathbf{u}$ is an eigenvector of S . Then $\mathbf{u}^\top S \mathbf{u} = \mathbf{u}^\top (\lambda \mathbf{u}) = \lambda$, so the maximum over eigenvectors is achieved at the largest eigenvalue, $\lambda = \lambda_1$, giving $\mathbf{u} = \mathbf{v}_1$. $\square$

Proposition 7.4. $PC_2 = \mathbf{v}_2$, the second eigenvector of S .

Proof. Maximize $\mathbf{u}^\top S \mathbf{u}$ subject to $\|\mathbf{u}\|^2 = 1$ and $\langle \mathbf{u}, \mathbf{v}_1 \rangle = 0$. The Lagrangian is $L(\mathbf{u}, \lambda, \mu) = \mathbf{u}^\top S \mathbf{u} - \lambda(\|\mathbf{u}\|^2 - 1) - 2\mu \langle \mathbf{u}, \mathbf{v}_1 \rangle$, and stationarity gives

$$S\mathbf{u} = \lambda \mathbf{u} + \mu \mathbf{v}_1.$$

Taking $\langle \mathbf{v}_1, \cdot \rangle$ of both sides: $\mathbf{v}_1^\top S \mathbf{u} = \lambda \langle \mathbf{u}, \mathbf{v}_1 \rangle + \mu \langle \mathbf{v}_1, \mathbf{v}_1 \rangle = \mu$ (using $\langle \mathbf{u}, \mathbf{v}_1 \rangle = 0$, $\|\mathbf{v}_1\| = 1$). But also $\mathbf{v}_1^\top S \mathbf{u} = (S\mathbf{v}_1)^\top \mathbf{u} = \lambda_1 \mathbf{v}_1^\top \mathbf{u} = 0$ (symmetry of S , $S\mathbf{v}_1 = \lambda_1 \mathbf{v}_1$, and the constraint again). So $\mu = 0$, and $S\mathbf{u} = \lambda \mathbf{u}$: $\mathbf{u}$ is again an eigenvector, now constrained to be orthogonal to $\mathbf{v}_1$, so the maximizing choice is the largest *remaining* eigenvalue, λ_2 , giving $\mathbf{u} = \mathbf{v}_2$. $\square$

Remark 7.5. The pattern continues: $PC_i = \mathbf{v}_i$, with $\lambda_i = \max\{\mathbf{u}^\top S \mathbf{u} : \|\mathbf{u}\| = 1, \mathbf{u} \perp \mathbf{v}_1, \dots, \mathbf{v}_{i-1}\}$. The general statement is a variational characterization of eigenvalues, the Courant–Fischer min-max theorem; we derive the base cases above and cite Courant–Fischer for the general argument.

Corollary 7.6 (PCA optimality is Eckart–Young). *The best rank-1 approximation of S (Frobenius norm) is $\lambda_1 \mathbf{v}_1 \mathbf{v}_1^\top$.*

Proof. $\|S - \lambda \mathbf{u} \mathbf{u}^\top\|^2 = \langle S, S \rangle - 2\lambda \langle S, \mathbf{u} \mathbf{u}^\top \rangle + \lambda^2 \langle \mathbf{u} \mathbf{u}^\top, \mathbf{u} \mathbf{u}^\top \rangle = \|S\|^2 - 2\lambda (\mathbf{u}^\top S \mathbf{u}) + \lambda^2$ for unit $\mathbf{u}$. Minimizing over λ : $\partial_\lambda(\dots) = -2\mathbf{u}^\top S \mathbf{u} + 2\lambda = 0 \Rightarrow \lambda = \mathbf{u}^\top S \mathbf{u}$, and substituting back leaves $-(\mathbf{u}^\top S \mathbf{u})^2$ to minimize, i.e. $(\mathbf{u}^\top S \mathbf{u})^2$ to maximize, i.e. (since $\mathbf{u}^\top S \mathbf{u} \geq 0$) $\mathbf{u}^\top S \mathbf{u}$ to maximize. By Proposition 7.3, $\mathbf{u} = \mathbf{v}_1$, $\lambda = \lambda_1$. $\square$

Corollary 7.6 is the $r = 1$ case of Theorem 6.4; PCA’s optimality for general r (maximizing total variance over r -dimensional subspaces) is the general case.

8 Perspectives on PCA

The same factorization $S \approx V_r \Lambda_r V_r^\top$ has multiple interpretations. Each is the starting point of a school of thought in data analysis.

1. **Parameter estimation.** Model: S has a low-rank structure, $S \approx V D V^\top$ for unknown $\mathbf{v}_i, d_i$; PCA estimates these parameters.
2. **Feature extraction.** $Z = X V_r \in \mathbb{R}^{n \times r}$: each row is a sample's coordinates in the r latent features. These $\mathbf{z}$ are useful downstream (e.g. to classify or cluster data).
3. **Dimensionality reduction / visualization.** Plotting the rows of $Z = X V_r$ (usually for $r = 2$) shows each data point in a low-dimensional space, revealing discrete and continuous structure in the p dimensions.

These are three facets of the field of *representation learning*: (1) parameter estimation under other statistical models; (2) feature extraction, e.g. via the learned representations inside a neural network or an autoencoder; (3) dimensionality reduction, including via nonlinear methods such as t-SNE or UMAP. PCA is the classical, linear instance of each.

Figures 4 and 5 are both pictures of PCA, for the example of approximating points in $\mathbb{R}^2$ by points in $\mathbb{R}^1$. Figure 5 is one you've probably seen before: the top PC is the line of best fit of the data in its least squares sense. Figure 4 is probably a picture of PCA you haven't seen before. It is the same problem viewed as projection to a sub-model of the PSD cone. Both are helpful perspectives when generalizing PCA.

9 The Singular Value Decomposition (SVD)

Here we examine the diagonalization and best approximation of the data matrix itself, $X \in \mathbb{R}^{n \times p}$.

Theorem 9.1 (Singular value decomposition). *Any $M \in \mathbb{R}^{n \times p}$ can be written $M = U D V^\top$, with $U \in \mathbb{R}^{n \times n}$ orthogonal, $V \in \mathbb{R}^{p \times p}$ orthogonal, and $D \in \mathbb{R}^{n \times p}$ diagonal (padded with zero rows or columns as needed) with non-negative entries $\sigma_1 \geq \dots \geq \sigma_{\min(n,p)} \geq 0$, the singular values.*

Remark 9.2 (Singular vector pairs). If $M = U D V^\top$, then $M \mathbf{v}_i = \sigma_i \mathbf{u}_i$ and $M^\top \mathbf{u}_i = \sigma_i \mathbf{v}_i$: $(\mathbf{u}_i, \mathbf{v}_i)$ is a singular vector pair with singular value σ_i .

Proof of Theorem 9.1. $M^\top M \in \mathbb{R}^{p \times p}$ is symmetric, and PSD since $\mathbf{v}^\top (M^\top M) \mathbf{v} = \|M \mathbf{v}\|^2 \geq 0$. By the spectral theorem, $M^\top M = V \Lambda V^\top$, V orthogonal, $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_p)$, $\lambda_i \geq 0$. Consider $W = M V \in \mathbb{R}^{n \times p}$: $W^\top W = V^\top M^\top M V = V^\top V \Lambda V^\top V = \Lambda$, diagonal. So the columns $\mathbf{w}_i$ of W are pairwise orthogonal, with $\|\mathbf{w}_i\|^2 = \lambda_i =: \sigma_i^2$. Set $\mathbf{u}_i = \mathbf{w}_i / \sigma_i$ where $\sigma_i > 0$ (extending to an orthonormal basis of $\mathbb{R}^n$ where $\sigma_i = 0$ or $p < n$), so that $W = U D$ with U orthogonal and $D = \text{diag}(\sigma_1, \dots)$. Then $M V = U D$, and since V is orthogonal, $M = U D V^\top$. $\square$

The SVD writes $M = \sum_i \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$. This can be viewed as a sum of modules: left singular vectors are per-sample scores, right singular vectors are per-variable loadings, and the singular value is a module's strength.

Remark 9.3 (PCA is the SVD of the data matrix). Since $S = \frac{1}{n} X^\top X$, applying Theorem 9.1 to the mean-centered $X = U D V^\top$ gives $X^\top X = V D^\top D V^\top$, matching S 's eigendecomposition with $\lambda_i = \sigma_i^2 / n$: the principal components are the right singular vectors of the mean-centered data matrix, and the PCA scores are $Z = X V = U D$.

Remark 9.4 (Ranks). The rank of a matrix is the number of non-zero singular values. The rank of a symmetric matrix is the number of non-zero eigenvalues. Given a symmetric matrix, its rank as a sum of symmetric rank one matrices $\mathbf{v}\mathbf{v}^\top$ equals its rank as a sum of not-necessarily-symmetric matrices $\mathbf{u}\mathbf{v}^\top$, since the number of non-zero eigenvectors and singular vectors coincide.

Theorem 9.5 (Eckart–Young, general case). *Let $M = UDV^\top \in \mathbb{R}^{n \times p}$ and let $M_r = U_r D_r V_r^\top$ be its truncation to the top r singular vector pairs, with their singular values. Then for any B with $\text{rank}(B) \leq r$, $\|M - B\| \geq \|M - M_r\|$.*

We do not reprove this (the argument mirrors Theorem 6.4 but is not identical, since M need not be symmetric); see Strang, *Linear Algebra and Learning from Data*.

10 Other Matrices to Diagonalize

Benzécri was a French mathematician and statistician. He is credited with expressing the sentiment that “all data analysis is matrix diagonalization; the crux of the matter is in finding the right matrix to diagonalize.” So far we have diagonalized covariance matrices and data matrices. Data analysis is full of other choices:

Matrix	Entry M_{ij}
Distance matrix	a distance between samples i and j
Contingency table	count of joint occurrences of outcome i (of one variable) and j (of another)
Indicator matrix	1 iff variable j occurs in sample i
Cross-covariance	$\text{Cov}(x_i, y_j)$, between two different variable sets
Correlation matrix	covariance, normalized by standard deviations
Moment matrix	$\mathbb{E}[x_i x_j]$ (no mean-centering)

Not every useful invariant of $\mathbf{x}$ is a matrix. The fourth cumulant $\kappa_4(\mathbf{x})$ is a $p \times p \times p \times p$ array (called the *kurtosis tensor*). Diagonalization of matrices is just the beginning of the story. The next module of the course is about tensors, such as the kurtosis tensor, and their decompositions.

References

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- R. Horn & C. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2012.
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